

# Successive h-indices using OpenAlex

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## Abstract

I apply Schubert’s successive h-index framework to OpenAlex, computing  $h_2$  for 22,980 institutions and  $h_3$  for 214 countries across 13.3 million researchers. The overall institution-level ranking is, in substance, a ranking of medical school size: Medicine accounts for 24% of the author pool and structurally dominates any university-wide aggregate. Field-specific decomposition is more informative: according to this metric, Wageningen University leads Agricultural & Biological Sciences and the University of Chinese Academy of Sciences leads Environmental Science, Materials Science, and Engineering; Carnegie Mellon leads Computer Science from 159th; Brazil holds two of the world’s top-six dental-research institutions by field  $h_2$ . I normalize for group size using Egghe’s Lotkaian scaling, which accounts for approximately 90% of variance in both  $h_2$  and  $h_3$ , and find that Italy and Japan tie on  $h_3$  despite a 6× difference in institutional scale.

**Keywords:** successive h-index, OpenAlex, institutional ranking, research assessment

**JEL Classification:** I23 , C80

**MSC Classification:** 62P99 , 62-07 , 62P25

## 1 Introduction

In 2007, András Schubert published a paper titled “Successive h-indices.” (2007) It begins:

Hirsch’s excessively discussed h-index generated a long row of interpretations, applications and derivatives within a surprisingly short period. Symptomatically, Hirsch’s paper in the *Proceedings of the National Academy*

of *Sciences of the United States of America* became the second highest cited paper of the issue in which it had been published. In this note, attention is called to an apparently so far overlooked option: h-indices themselves may form the basis of a successive series of h-indices.

Given, e.g., the h-indices of the researchers of an institute (which indices will be denoted by  $h_1$  in what follows), an index,  $h_2$ , of the institute can be determined: the institute has an index  $h_2$  if  $h_2$  of its  $N$  researchers have an  $h_1$ -index of at least  $h_2$  each, and the other  $N - h_2$  researchers have  $h_1$ -indices lower than  $h_2$  each. The succession can then be continued, e.g., for networks of institutions or countries or other higher levels of aggregation.

Since no sufficient data for testing the concept on the researcher-institute-country hierarchy were available, another example was chosen.

Fifteen years later, OpenAlex (Priem, Piwowar, & Orr, 2022) launched a FOSS bibliographic catalogue complete with citation indexing, authors, fields, and institutions. Recent evaluations have found it to be a broadly suitable substitute for proprietary sources such as Scopus and Web of Science at the institutional level (Alperin, Portenoy, Demes, Larivière, & Haustein, 2024; Culbert et al., 2025), and I take it here as the dataset for which Schubert wished in 2007.

The idea of applying the h-index hierarchically at the researcher-institution level was proposed independently by Prathap (2006) in a one-page note less than a month before Schubert's paper. Schubert's distinctive contribution was to recognize that the construction extends to an arbitrary succession of levels, opening the path to  $h_3$  and beyond. In the years between, successive h-indices attracted modest but sustained attention. Egghe (2008) gave the concept a theoretical footing, showing how the exponent of the underlying Lotka distribution propagates through each successive aggregation step. Several case studies followed shortly after: Ruane and Tol (2008) computed a rational (non-integer) successive h-index across economics departments in Ireland; Arencibia-Jorge, Barrios-Almaguer, Fernández-Hernández, and Carvajal-Espino (2008) applied Schubert's exact researcher-department-institution hierarchy to a Cuban research institute using Web of Science data; and Arencibia-Jorge and Rousseau (2009) applied Prathap's framing to twelve Cuban brain-research institutions. None of this prior work, however, has had access to a comprehensive, open, institution-labeled snapshot of the kind OpenAlex now provides.

Schubert suggested that  $h_2$  of institutes would be more interesting than that of journals; Section 2 describes how I compute it at scale, and Section 3 presents the results.

## 2 Methodology

### 2.1 Data source and filtering

The underlying data is OpenAlex's public authors snapshot (Priem et al., 2022), distributed as Parquet on `s3://openalex/data/parquet/authors/` (`us-east-1`, no credentials required) and partitioned into over 1,600 daily `updated_date=` directories

totalling about 53 GB. Only four columns are read from each remote file (`id`, `h_index`, `affiliations`, `topics`), and an author is retained only if all of the following hold:

- `summary_stats.h_index` is non-null and greater than zero;
- at least one entry in `affiliations` has `institution.type = 'education'`;
- at least one entry in `topics` carries a non-null field classification.

Applying these filters to the full snapshot yields **30.0 million** authors across OpenAlex’s 26 fields. One limitation carried over from the source data: an author’s `h_index` is computed by OpenAlex from all works meeting its citation criteria regardless of retraction status, which OpenAlex records inconsistently (Hauschke & Nazarovets, 2025); a small, unquantified fraction of  $h_1$  values in the dataset may therefore be inflated by since-retracted work.

## 2.2 Author-level assignment

Each retained author is reduced to a single row: one institution and one field. For institution assignment, rather than reading the institution from the author’s affiliation list in the snapshot, I derive it from the OpenAlex works snapshot (Parquet files at `s3://openalex/data/parquet/works/`, no credentials required), because the `affiliations[]` field in the author record records only which calendar years an institution appears on the author’s papers, not on how many. An authorship in the works snapshot credits education institutions as follows (hereafter called the “rollup” rule, applied by default): every institution with `type = 'education'` listed on the authorship is credited directly; additionally, for every non-education institution listed, its education-type ancestors in OpenAlex’s institution `lineage` are credited (so a Jet Propulsion Laboratory authorship credits Caltech); education institutions are never rolled up to their own parents, so a UC Berkeley paper does not also credit the University of California System. An author’s primary institution is then the education institution credited on the most of their authorships in the last five publishing years (the author’s most recent publication year and the four calendar years before it). Two thresholds must be met for an assignment to be made: the institution must be credited on at least two works in the window, and on at least 10% of the window’s works that credit *any* education institution. The second threshold uses the education-crediting works as its denominator (rather than all the author’s works) so that hospital- and institute-based faculty whose papers mostly list only their hospital are not penalised when the hospital has an education-type parent through the rollup rule. Among authors who meet the thresholds, ties go first to the institution seen most recently in the window, then to the institution appearing most often across the author’s whole career, then to the numerically smallest OpenAlex institution identifier. Of 126.6 million unique authors appearing in the works snapshot, 41.8 million (33.0%) have at least one education institution credited in their five-year window; 25.9 million fail the minimum-papers threshold, and 2,326 fail the share threshold, leaving 15.9 million with a primary institution assigned. Joining these with the 30.0 million authors in the snapshot who have at least one education affiliation, and at least one topic classification yields the 13.3 million authors across 22,980 institutions that form the basis for all rankings in this paper.

Field assignment works the same way but is weighted. OpenAlex tags each author with a set of topics, each carrying a `count` (roughly, the number of the author’s works classified under that topic) and a parent field. I sum `count` within each field and assign the author to whichever field carries the largest total weight: their modal field by publication volume, not simply their single most granular topic. The result is an `authors` table with one row per author: (`author_id`, `h_index`, `institution_id`, `institution_name`, `field`, `field_name`).

Two structural features of this procedure limit the impact of field classification errors. First, the paper operates at the *field* level (26 fields) rather than at the finer topic level ( $\sim 4,500$  topics); OpenAlex’s topic classifier fine-tunes a multilingual BERT model on title, abstract, and citation data and achieves a top-1 accuracy of 0.53 (OpenAlex, 2024), but many topic-level errors stay within the same parent field, so field-level accuracy is *plausibly higher, though this is an inference from the structure of the classification hierarchy rather than a directly measured field-level accuracy figure: no field-level validation of the classifier is presented here*. Second, an author’s field is determined by the mode of their *entire publication record*, not by a single work: a researcher would need a substantial fraction of their career output systematically misclassified to a different field before their assignment would change.

### 2.3 The h-index and its successive generalization

Hirsch’s (2005) original index, here  $h_1$ , is computed by OpenAlex per author from citation counts and taken as given. Schubert’s generalization (2007) lifts the same definition up a level of aggregation: a group of  $N$  members, each already carrying an  $h_{k-1}$  index, has an index  $h_k = n$  if  $n$  of its members have  $h_{k-1} \geq n$  and the remaining  $N - n$  members have  $h_{k-1} < n$ :

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**Algorithm 1** Successive h-index of a group  $G$ , given a value  $v(m)$  for each member  $m \in G$

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- 1: Sort  $G$  by  $v$  descending; let  $r(m)$  be  $m$ ’s 1-indexed rank in that order
  - 2: **return**  $\max\{r(m) : v(m) \geq r(m), m \in G\}$
- 

I use this rank-based procedure, implemented as a window function in SQL ranking rows within each group and taking the largest rank whose value has not yet fallen below it, for every successive index in this paper:  $h_2$  of an institution over the  $h_1$  of its authors, and  $h_3$  of a country over the  $h_2$  of its institutions.

### 2.4 $h_2$ : institution and institution $\times$ field

$h_2$  is computed twice, over different groupings of the same author-level data. The field-specific table groups authors by (institution, field) and yields 288,300 pairs; institutions are matched on their numeric identifier only; institution and field *names* are resolved afterward via `arg_max`, so that spelling or formatting variants of the same name across snapshot partitions cannot silently split one institution into several groups. The institution-overall table ignores field entirely and applies the same procedure across

all of an institution’s authors regardless of discipline, yielding one  $h_2$  per institution (22,980 rows with a named institution).

## 2.5 $h_3$ : country level

Institutions are mapped to countries using the `country_code` field of OpenAlex’s institutions snapshot (also read directly from S3, joined on institution ID).  $h_3$  is then computed with the same rank-based procedure, once globally per country over the  $h_2$  values of its institutions and once per (country, field) over the field-specific  $h_2$  values, producing one file per field.

## 2.6 Efficiency

Mryglod, Holovatch, and Kenna (2022) showed that group-level h-indices grow with group size even when per-member quality is held constant. They were not studying successive h indices, but a similar argument makes raw  $h_2$  and  $h_3$  values unsuitable for comparing institutions or countries of unequal scale.

Egghe (2008) applied a Lotkaian framework to successive h indices. Suppose the article–citation frequency function follows Lotka’s law with exponent  $\alpha_0 > 1$ , and the author–article frequency function follows Lotka’s law with exponent  $\alpha_1 > 1$ . Egghe showed that the distribution of  $h_1$  values across authors is itself Lotkaian with compound exponent  $\beta_1 = \alpha_0\alpha_1$ . Applying the h-index formula for Lotkaian systems, the institution-level index satisfies

$$h_2 = S^{1/\beta_1} \quad (1)$$

where S is the number of authors at the institution. An analogous argument at the next level gives the distribution of  $h_2$  values across institutions a Lotkaian exponent  $\beta_2 = \alpha_0\alpha_1\alpha_2$ , where  $\alpha_2$  is the Lotka exponent of the institution–author frequency function, so that

$$h_3 = R^{1/\beta_2} \quad (2)$$

where R is the number of institutions in the country. I define size-normalized efficiency metrics by dividing out these predicted scalings:

$$\varepsilon_2 = \frac{h_2}{S^{1/\beta_1}} \quad (3)$$

$$\varepsilon_3 = \frac{h_3}{R^{1/\beta_2}} \quad (4)$$

An institution with  $\varepsilon_2 > 1$  outperforms the Lotkaian prediction for its headcount; one with  $\varepsilon_2 < 1$  underperforms it.

The exponents  $\beta_1$  and  $\beta_2$  are estimated directly from the data rather than assembled from separately fit  $\alpha_0, \alpha_1, \alpha_2$ . I apply discrete maximum-likelihood tail estimation following Clauset, Shalizi, and Newman (2009), with  $x_{\min}$  chosen to minimize the Kolmogorov–Smirnov distance between the empirical and fitted CDFs, to the distributions of  $h_1$  across all authors and  $h_2$  across all institutions, reading  $\beta_1$  and  $\beta_2$  as their fitted Lotka exponents. I then fit  $\alpha_0, \alpha_1$ , and  $\alpha_2$  individually:  $\alpha_0$  from a stratified sample of per-article citation counts drawn from the OpenAlex works snapshot,  $\alpha_1$  from

per-author work counts, and  $\alpha_2$  from per-institution author counts. Finally, I verify that  $\alpha_0\alpha_1 \approx \beta_1$  and  $\alpha_0\alpha_1\alpha_2 \approx \beta_2$  as a cross-check that the compounding-exponent structure of Egghe’s model holds in the OpenAlex dataset.

## 2.7 Supplementary metrics

Several secondary statistics are derived from the  $h_2$  and  $h_3$  tables to probe what the headline numbers obscure:

- **Gini coefficient**, computed on the within-field distribution of institutional  $h_2$ , quantifies how concentrated institutional  $h_2$  is within a field (Gini near 1) or how broadly distributed (Gini near 0).
- **Breadth score** counts, per institution, how many of the 26 fields it ranks in the global top 10, top 50, and top 100.
- **Non-biomedical ranking** recomputes institution-level  $h_2$  after excluding authors whose modal field is Medicine, Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, or Health Professions, to test whether the overall ranking is, in substance, a ranking of medical school size.

## 2.8 Manual cross-validation

Because errors in OpenAlex’s underlying author disambiguation or citation counts would propagate directly into every successive index, I manually spot-check a targeted, non-random sample of authors rather than a random one, chosen at the points where an error would be both most likely and most consequential: authors sitting exactly at an institution’s  $h_2$  threshold, where a  $\pm 1$  error in a single  $h_1$  could change the institution’s rank; top authors in fields where OpenAlex’s topic classification is known to be thinner (e.g., Arts and Humanities); top authors at institutions with non-Anglophone naming conventions, where author-name disambiguation is more error-prone and where OpenAlex’s coverage is known to diverge unevenly from other databases by country, most severely for China (Zheng, Miao, Bu, & Larivière, 2025), including institutions suspected of affiliation-swapping for grant purposes (Bhattacharjee, 2011); and the small number of authors with the highest  $h_1$  globally, as a sanity check at the extreme of the distribution. For each candidate, I resolve the author’s OpenAlex profile through the public API and compare it by hand against Web of Science, Scopus, and Google Scholar. This design is deliberately adversarial: it is built to surface errors, not to estimate how frequently they occur across the full population, so any outlier rate found in it bounds the qualitative failure mode (which authors, and by how much) rather than the population-wide error rate.

## 2.9 Sensitivity to disambiguation error

Because the cross-validation sample above cannot itself bound a population-wide error rate, I separately ask how much author-disambiguation error the paper’s conclusions could tolerate before changing using a Monte Carlo sensitivity analysis run directly on the full authors table. At a population-wide error rate  $p$ , I flag each author independently with probability  $p$ . For authors affiliated with institutions in

China, South Korea, Iran, or Saudi Arabia, I flag each author with probability  $\min(1, 3p)$ , reflecting the cross-validation finding that disambiguation errors concentrate in non-Anglophone, common-name strata. A flagged author’s  $h_1$  is perturbed by a ratio drawn uniformly from the four ratios observed among the cross-validation outliers (103/57, 45/11, 215/115, 139/41), with equal probability of upward inflation ( $\times$  ratio, simulating a merged-identity error) or downward deflation ( $\times$  ratio<sup>-1</sup>, simulating a split-identity error). I recompute institution-level  $h_2$ , country-level  $h_3$ , and the field-specific  $h_2$  rankings, and compare each to the unperturbed baseline via Spearman rank correlation and top-set overlap. Rather than picking a single rate to call a “worst case,” I run **fifty** independent trials at each of **seven** rates,  $p \in \{1\%, 5\%, 10\%, 13\%, 20\%, 40\%, 80\%\}$ , tracing out how ranking stability responds to the assumed amount of error from a plausible rate up to a deliberately implausible one.

## 2.10 Robustness to field assignment

The single-field assignment described in Subsection 2.2 asks “which field accounts for the largest share of an author’s publication volume?” and assigns each author exclusively to that field. To test whether field-specific  $h_2$  rankings are stable under an alternative that allows multidisciplinary authors to count in multiple fields simultaneously, I re-run the field-specific  $h_2$  computation after expanding each author’s field membership: at a threshold  $\tau$ , an author is assigned to every field in which their OpenAlex topic-weight share is at least  $\tau$  (where topic-weight is the same count-weighted field sum used in the original assignment). I test two thresholds,  $\tau \in \{20\%, 33\%\}$ , spanning from liberal (an author need contribute only one fifth of their output to count in a field) to moderate (one third or more). For each threshold I recompute  $h_2$  for every (institution, field) pair in the expanded author roster, using the same rank-based algorithm as the original, and compare to the baseline via Spearman rank correlation on shared (institution, field) pairs and by checking whether each field’s leading institution remains #1.

## 2.11 Robustness to institution assignment

Two classes of robustness test bear on the primary-institution assignment. The first asks whether the single-institution constraint itself distorts results. The primary rule credits each author to the one institution with the most works in their five-year window; a *multi-institution* variant instead credits each author’s full  $h_1$  to every institution meeting both thresholds in that window. This is the most natural analog of a “fractional” treatment given that  $h$ -indices are integer-valued, and directly tests whether multiply-affiliated researchers distort the single-institution rankings.

The second class asks how sensitive the results are to the three independently variable choices *within* the works-based rule: (1) whether to roll up non-education institutions to their education-type ancestors in the OpenAlex lineage (the “rollup” rule); (2) whether the 10% institution-share threshold uses education-crediting works or all of the author’s window works as its denominator; and (3) whether to use the works-based assignment at all, versus reading the most recent education institution directly

from the author snapshot. I test each choice independently, producing a *no-rollup* variant that suppresses ancestor credit (so a Jet Propulsion Laboratory authorship no longer credits Caltech); a *share-of-all* variant that computes the share over all window works rather than only education-crediting ones; and a *most-recent-snapshot* variant that assigns each author to the education institution appearing most recently in the snapshot, breaking ties by the most frequent career-wide appearance and then by smallest OpenAlex identifier. For all four variants, I recompute institution-level  $h_2$  and country-level  $h_3$  and compare against the paper’s default assignment via Spearman rank correlation and top-10 overlap.

## 2.12 External validation against Scimago Institutions Rankings

To assess whether  $\varepsilon_2$  captures genuine research quality rather than a residual size artefact, I compare raw  $h_2$  and  $\varepsilon_2$  against the Scimago Institutions Rankings (SIR) 2026 “Overall Rank” export (SCImago Research Group, 2026), a Scopus-derived composite ranking (Research 50%, Innovation 30%, Societal 20% sub-scores). I compute Spearman  $\rho$  between SIR’s Global Rank and both the raw  $h_2$  rank and the  $\varepsilon_2$  efficiency rank and repeat the comparison restricted to SIR’s “Universities” sector only, excluding the ministries, hospitals, and companies that SIR also ranks, to check whether any mismatch is an artefact of comparing against a mixed-sector list.

## 2.13 Software and reproducibility

All analyses were implemented in Python 3.10 (NumPy, pandas, geopandas, matplotlib, duckdb, powerlaw, scipy). The data fetching, filtering, and processing are fully reproducible from the public code repository at <https://github.com/rupumped/successive-h-indices>.

# 3 Results

## 3.1 Manual cross-validation

Of the 30 authors sampled under the scheme described in Subsection 2.8, Table 1 summarizes agreement between OpenAlex’s `h_index` and the corresponding value I located by hand in Web of Science, Scopus, and Google Scholar. Coverage itself varies sharply: Scopus returned a value for 26 of the 30 authors, Google Scholar for 19, and Web of Science for 16.

**Table 1** Agreement between OpenAlex and each external source, over authors for which that source returned a value. Diff is OpenAlex minus the external value.

| Source         | $n$   | Mean diff | Mean abs. diff | Worst case         |
|----------------|-------|-----------|----------------|--------------------|
| Web of Science | 16/30 | +24.5     | 25.7           | Lei Jiang (+100)   |
| Scopus         | 26/30 | +8.0      | 12.4           | Donald Small (+46) |
| Google Scholar | 19/30 | −22.7     | 30.7           | A. Jafari (−85)    |

Scopus tracks OpenAlex most closely, both in coverage and in magnitude of disagreement. Google Scholar is systematically higher than OpenAlex, consistent with its broader crawl of preprints, theses, and other grey literature and its comparative permissiveness toward self-citation; this ordering (Google Scholar > Scopus > OpenAlex > Web of Science) reproduces cleanly among the six global top- $h_1$  sanity-check authors, none of whom show a discrepancy exceeding 27% of their OpenAlex value in either direction.

Four authors, however, show disagreement well outside this range. I read this as a signature of author-disambiguation failure (multiple distinct people merged under one OpenAlex author ID) rather than of genuine cross-database disagreement:

- **Donald Small** (Johns Hopkins):  $h_1^{\text{OpenAlex}} = 103$  against Scopus = 57 and Google Scholar = 59.
- **Karen S. Calhoun** (University of Michigan):  $h_1^{\text{OpenAlex}} = 45$  against Scopus = 11 and Google Scholar = 13.
- **Lei Jiang** (University of Science and Technology of China):  $h_1^{\text{OpenAlex}} = 215$  against Web of Science = 115, consistent with documented China-specific coverage discrepancies in OpenAlex (Zheng et al., 2025).
- **A. Jafari** (Sharif University of Technology):  $h_1^{\text{OpenAlex}} = 139$ , but Web of Science, Scopus, and Google Scholar report 41, 133, and 224 respectively.

All four fall in the non-Anglophone-institution or common-name strata of the sample, matching the concern raised in Subsection 2.8 that disambiguation is more error-prone for exactly these authors. Because such merges can only inflate an author’s apparent  $h_1$ , and  $h_1$  feeds directly into  $h_2$  and  $h_3$ , a handful of merged identities at the right institution could measurably distort an institution- or country-level ranking; I flag this as an open limitation rather than one this sample can size definitively. This sample is deliberately adversarial, chosen to find errors, not to estimate how often they occur. The 4/30 (13%) outlier fraction should not be read as an estimate of how much of the full dataset is affected. The same pattern, disambiguation precision concentrated lower for common names and non-Anglophone institutions, appears in an independent benchmark of 2,700 scholars sampled systematically across three regions (Zhao & Chen, 2025), corroborating that this is a geographically and onomastically concentrated failure mode rather than an artifact of this particular sample. OpenAlex’s maintainers report an ongoing correction effort targeting exactly this kind of over-merging, with corresponding-author identification precision rising from 0.60 to 0.92 (F1 0.72 to 0.90) after a 2026 rework of the underlying author-matching model (Demes, 2026). Subsection 3.12 quantifies how much a population-wide error rate at or below the adversarial sample’s raw rate would actually move this paper’s rankings.

The four Arts and Humanities threshold authors sampled (Gillian Ramchand and G. Herbert Fowler at Oxford; D.H. Mellor and James Raven at Cambridge) illustrate the range of under-coverage across major citation databases: Ramchand and Fowler returned no value in any of the three external sources, while Mellor (OA  $h_1 = 28$ ) and Raven (OA  $h_1 = 24$ ) appeared in Scopus at roughly half their OpenAlex values ( $h_1 = 13$  and 12, respectively); for Raven, Google Scholar (26) brackets the OpenAlex

value closely. This is consistent with the well-documented weaker coverage of book-based humanities scholarship in all major citation databases (Nederhof, 2006) and suggests that h-index comparisons in these fields should be read with correspondingly less confidence than in the sciences.

### 3.2 The overall $h_2$ ranking

Table 2 lists the ten institutions with the highest institution-level  $h_2$ , computed over all 22,980 institutions and 13.3 million authors described in Section 2.

**Table 2** Top 10 institutions by overall  $h_2$ .

| Institution                           | $h_2$ | Authors |
|---------------------------------------|-------|---------|
| Harvard University                    | 135   | 47,279  |
| Stanford University                   | 113   | 36,976  |
| University of Antwerp                 | 106   | 7,207   |
| University of California San Diego    | 106   | 24,953  |
| University of Cambridge               | 106   | 24,810  |
| University of Oxford                  | 105   | 28,577  |
| University of Washington              | 104   | 30,392  |
| Johns Hopkins University              | 102   | 33,329  |
| Massachusetts Institute of Technology | 101   | 21,527  |
| University College London             | 101   | 33,413  |

This broadly tracks the US News and World Report ranking, where Harvard also places 1st, Stanford 3rd, and Cambridge 5th. As I show below, however, the overall ranking is a considerably less interesting object than it first appears.

### 3.3 Field-level heterogeneity

Splitting  $h_2$  by (institution, field) reveals that the overall ranking is deceptive: several institutions that never come close to the top of Table 2 rank first globally in their field by this measure. Table 3 lists a sample of these cases.

University of Antwerp is the counterexample that proves the rule holds both ways: it is #3 overall ( $h_2 = 106$ ) and #1 in Physics and Astronomy, a broad powerhouse that also ranks first in a specific, non-biomedical field. Universidade Estadual de Campinas (UNICAMP) places 6th in Dentistry behind São Paulo, placing Brazil among the top six globally in Dentistry by this measure despite the country appearing nowhere near the top of any general-purpose ranking (Vaccaro, Sánchez-Núñez, & Witt-Rodríguez, 2022). China’s leading field-level  $h_2$  positions in applied sciences are concentrated in the University of Chinese Academy of Sciences (UCAS, rank 24 overall,  $h_2=95$ ), which leads Environmental Science, Materials Science, Engineering, and Chemical Engineering by field  $h_2$ , while its sister institution, the University of Science and Technology of China (USTC, rank 66,  $h_2=84$ ), leads Energy.

Not every field supports this kind of comparison equally well. Decision Sciences is the thinnest field in the dataset: Stanford’s global #1 is  $h_2 = 20$ , built on only 129

**Table 3** Selected field leaders that rank well outside the overall top 10. “Overall” columns give the institution’s rank and  $h_2$  in Table 2’s full ranking; “Field  $h_2$ ” is the institution’s  $h_2$  restricted to that field.

| Field                                    | Leader                               | Ov. rank | Ov. $h_2$ | Field $h_2$ |
|------------------------------------------|--------------------------------------|----------|-----------|-------------|
| Agricultural & Biological Sciences       | Wageningen                           | 148      | 73        | 58          |
| Environmental Science                    | Univ. of Chinese Academy of Sciences | 24       | 95        | 60          |
| Pharmacology, Toxicology & Pharmaceutics | Shenyang Pharmaceutical University   | 957      | 40        | 26          |
| Computer Science                         | Carnegie Mellon University           | 159      | 71        | 56          |
| Materials Science                        | Univ. of Chinese Academy of Sciences | 24       | 95        | 71          |
| Engineering                              | Univ. of Chinese Academy of Sciences | 24       | 95        | 71          |
| Energy                                   | USTC                                 | 66       | 84        | 51          |
| Chemical Engineering                     | Univ. of Chinese Academy of Sciences | 24       | 95        | 30          |
| Dentistry                                | Universidade de São Paulo            | 226      | 66        | 37          |

Stanford authors against 56,987 authors field-wide, versus Medicine’s  $h_2 = 118$  built on 3.23 million authors field-wide.  $h_2$  is not meaningfully comparable across fields of such different size, only within a field, and Decision Sciences in particular may be too sparse an OpenAlex topic category for its ranking to carry much signal.

### 3.4 Robustness of field-specific rankings to multi-field assignment

The single-field assignment described in Subsection 2.2 asks which field accounts for the plurality of an author’s publication volume, and assigns them exclusively there. To test how much this choice moves the field-level rankings, I re-run the field-specific  $h_2$  computation after allowing each author to count in every field for which their OpenAlex topic-weight share is at least  $\tau$  in accordance with Subsection 2.10.

At  $\tau = 33\%$  (an author must contribute at least one third of their output to count in a field), the (institution, field) pair count rises by 16.0% (229,309  $\rightarrow$  266,002 pairs). The Spearman rank correlation between original and multi-field  $h_2$  values, computed on the inner join of shared pairs, is  $\rho = 0.960$ : field rankings are highly stable. Of

the 28 fields’ leading institutions (26 fields, but Social Sciences and Business, Management and Accounting each have two institutions tied for the top  $h_2$  value, yielding 28 leader entries), 23 are unchanged. The Wageningen Ag&Bio leadership (Table 3) survives, Carnegie Mellon’s Computer Science lead survives (tied with MIT,  $h_2 = 56$ ), and the University of Chinese Academy of Sciences (UCAS) Environmental Science lead ( $h_2 = 62$ , baseline  $h_2 = 60$ ) also survives. The two methods answer different questions. The single-field assignment identifies institutions where Environmental Science is researchers’ *primary* field; the multi-field assignment identifies institutions with the most researchers for whom it is a *substantial* field.

At  $\tau = 20\%$  (a more aggressive threshold admitting all authors with one fifth or more of their output in a field), the pair count rises by 39.6% (229,309 to 320,067) and  $\rho = 0.901$ ; 21 of 28 field leaders survive. Carnegie Mellon’s Computer Science lead also falls at this threshold, with Stanford leading instead ( $h_2 = 62$ ), consistent with the same mechanism of Stanford having a larger population of researchers with at least 20% CS output, even if CS is not their modal field. These findings confirm that the overall field ranking structure is broadly stable under multi-field assignment but that the field-leadership claims for smaller, highly specialised institutions depend on how narrowly “counts in this field” is defined.

### 3.5 Robustness of institutional rankings to attribution method

Table 4 reports rank-correlation statistics for the four sensitivity variants described in Subsection 2.11, each compared against the paper’s default primary-institution assignment. Institution-level  $h_2$  Spearman rank correlations  $\rho$  are computed over institutions shared between the default and variant rosters; country-level  $h_3$  correlations use the fixed institution-to-country mapping.

**Table 4** Sensitivity of institution  $h_2$  and country  $h_3$  rankings to four alternative attribution methods, each compared against the default primary-institution assignment. Spearman  $\rho$  is computed over shared institutions (for  $h_2$ ) or shared countries (for  $h_3$ ); top-10 overlap counts institutions/countries common to both top-10 lists.

| Attribution variant  | $h_2$ $\rho$ | $h_2$ top-10 | $h_3$ $\rho$ | $h_3$ top-10 |
|----------------------|--------------|--------------|--------------|--------------|
| Multi-institution    | 0.9391       | 7/10         | 0.9954       | 9/10         |
| No rollup            | 0.9985       | 9/10         | 0.9976       | 10/10        |
| Share of all works   | 0.9986       | 9/10         | 0.9973       | 10/10        |
| Most-recent snapshot | 0.9074       | 7/10         | 0.9906       | 9/10         |

The no-rollup and share-of-all variants are nearly indistinguishable from the default: institution  $h_2$  Spearman correlations of 0.998–0.999 and top-10 overlaps of 9/10 confirm that neither the rollup rule nor the denominator choice for the 10% share threshold materially shifts the rankings.

The most-recent-snapshot variant, which reverts to reading each author’s primary institution directly from the OpenAlex author snapshot rather than from the works-based five-year window, produces a lower institution  $h_2$  Spearman correlation of 0.907. Seven of the ten top-ranked institutions are shared with the default top-10, but the three that differ include institutions whose apparent performance under the most-recent-snapshot rule reflects single-paper secondary affiliations rather than sustained research presence, the pattern the primary-institution rule was designed to exclude. Country  $h_3$  is more stable ( $\rho = 0.991$ , 9/10 top-10 overlap), consistent with country aggregation smoothing institution-level noise. These results confirm that the choice between snapshot-based and works-based attribution is the dominant source of sensitivity in this paper’s rankings, while choices within the works-based approach (rollup, denominator) are comparatively minor.

Rather than assigning each author to a single institution, the multi-institution variant credits each author’s full  $h_1$  to every institution meeting both thresholds in their five-year window. Of the 13.3 million qualifying authors, 18.0% (2,858,984) qualify at two or more institutions under this rule, producing 17.2 million (author, institution) attributions. Institution  $h_2$  Spearman correlation is  $\rho = 0.939$  (7/10 top-10 overlap); country  $h_3$  is  $\rho = 0.995$  (9/10 top-10 overlap). The three institutions entering the multi-institution top-10 that were not in the default top-10 (the University of Adelaide, Université Paris-Saclay, and Université Paris Cité) are large research universities whose researchers commonly hold dual appointments with national research agencies (CNRS, 2026; Inserm, 2021), so they gain more authors under a multi-institution rule. The overall structure of the rankings is broadly stable at both the institution and country level.

### 3.6 Concentration within fields

Table 5 reports, for every field, the Gini coefficient of institutional  $h_2$  defined in Section 2.7, together with the field’s institution count, leading (maximum)  $h_2$ , and mean  $h_2$ .

Physics and Astronomy is the most unequal field: University of Antwerp’s  $h_2 = 104$  is 16.2 times the field mean of 6.4. Decision Sciences is the most evenly distributed (Gini  $\approx 0.43$ ).

### 3.7 The overall ranking as an implicit biomedical ranking

Medicine alone accounts for 3.23 million of the dataset’s 13.3 million authors (24%), and, together with Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, and Health Professions, the five core biomedical fields dwarf every other field’s author pool. Table 6 recomputes institution-level  $h_2$  after excluding authors in these five fields entirely to test whether the overall ranking in Table 2 is, in substance, a ranking of medical school size.

The University of Antwerp takes first place without biomedical authors (non-bio  $h_2=105$ , overall rank 3), followed by Harvard and Stanford (both non-bio  $h_2=100$ , overall ranks 1 and 2 respectively). Antwerp’s top position reflects high  $h$  indices across physics, chemistry, and life-sciences disciplines outside the core biomedical fields. The

**Table 5** Gini coefficient of institutional  $h_2$  within each field, sorted from most to least concentrated.

| Field                                      | Institutions | Max $h_2$ | Mean $h_2$ | Gini  |
|--------------------------------------------|--------------|-----------|------------|-------|
| Physics and Astronomy                      | 9,054        | 104       | 6.42       | 0.619 |
| Biochemistry, Genetics & Molecular Biology | 11,181       | 100       | 6.88       | 0.590 |
| Neuroscience                               | 7,808        | 62        | 4.94       | 0.581 |
| Medicine                                   | 16,228       | 118       | 7.40       | 0.579 |
| Earth and Planetary Sciences               | 7,787        | 58        | 4.74       | 0.573 |
| Immunology and Microbiology                | 6,095        | 63        | 4.46       | 0.568 |
| Materials Science                          | 10,008       | 71        | 6.75       | 0.562 |
| Environmental Science                      | 12,793       | 60        | 5.58       | 0.557 |
| Engineering                                | 14,486       | 71        | 6.78       | 0.556 |
| Agricultural and Biological Sciences       | 12,877       | 58        | 5.45       | 0.535 |
| Chemistry                                  | 8,463        | 50        | 5.35       | 0.535 |
| Psychology                                 | 11,486       | 56        | 3.83       | 0.530 |
| Dentistry                                  | 4,712        | 37        | 3.65       | 0.527 |
| Computer Science                           | 14,392       | 56        | 5.06       | 0.519 |
| Energy                                     | 6,246        | 51        | 3.92       | 0.514 |
| Mathematics                                | 8,046        | 35        | 3.74       | 0.490 |
| Social Sciences                            | 17,646       | 46        | 4.15       | 0.488 |
| Economics, Econometrics and Finance        | 10,238       | 54        | 3.47       | 0.486 |
| Business, Management and Accounting        | 11,909       | 32        | 3.77       | 0.485 |
| Health Professions                         | 10,317       | 40        | 3.05       | 0.479 |
| Veterinary                                 | 2,162        | 26        | 2.52       | 0.468 |
| Pharmacology, Toxicology and Pharmaceutics | 4,392        | 26        | 2.96       | 0.456 |
| Arts and Humanities                        | 10,759       | 28        | 2.70       | 0.451 |
| Chemical Engineering                       | 3,486        | 30        | 2.88       | 0.443 |
| Nursing                                    | 5,333        | 32        | 2.63       | 0.435 |
| Decision Sciences                          | 6,961        | 20        | 2.63       | 0.432 |

table reshuffles substantially: Paris-Saclay, Berkeley, UCAS, Caltech, and ETH Zurich all rise 10–26 places once Medicine and its four sibling fields are set aside, while Cambridge falls 6 places and Oxford 8. The University of Chinese Academy of Sciences is 6th non-biomedical (overall 24th), and The University of Adelaide enters the top 10 (10th non-bio, 34th overall). Because medical fields have by far the largest, most-cited author pools, they structurally dominate any university-wide aggregate; an institution elite in non-medical fields, such as USTC or Carnegie Mellon, can never reach the top of the overall list no matter how good it is, simply because it lacks a large medical school. The field-level and non-biomedical breakdowns matter more than the unrestricted university-level number.

### 3.8 Breadth versus depth

Table 3 showed institutions that lead a single field while sitting far outside the overall top 100. Table 7 shows the opposite pattern: institutions whose strength is broad

**Table 6** Top 10 institutions by non-biomedical  $h_2$ , alongside their position in the unrestricted overall ranking.  $\Delta\text{rank}$  is overall rank minus non-biomedical rank; positive means the institution moves up once biomedical fields are excluded.

| Institution                           | Rank | $h_2$ | Overall rank | Overall $h_2$ | $\Delta\text{rank}$ |
|---------------------------------------|------|-------|--------------|---------------|---------------------|
| University of Antwerp                 | 1    | 105   | 3            | 106           | +2                  |
| Harvard University                    | 2    | 100   | 1            | 135           | -1                  |
| Stanford University                   | 3    | 100   | 2            | 113           | -1                  |
| Université Paris-Saclay               | 4    | 95    | 14           | 100           | +10                 |
| University of California, Berkeley    | 5    | 94    | 16           | 99            | +11                 |
| Univ. of Chinese Academy of Sciences  | 6    | 93    | 24           | 95            | +18                 |
| Massachusetts Institute of Technology | 7    | 91    | 9            | 101           | +2                  |
| California Institute of Technology    | 8    | 90    | 26           | 94            | +18                 |
| ETH Zurich                            | 9    | 88    | 35           | 90            | +26                 |
| The University of Adelaide            | 10   | 88    | 34           | 88            | +24                 |

rather than deep, measured by how many of the 26 fields they place in the global top 50.

**Table 7** The nine institutions with the most top-50 field placements out of 26 fields; ties broken by top-10 field placements.

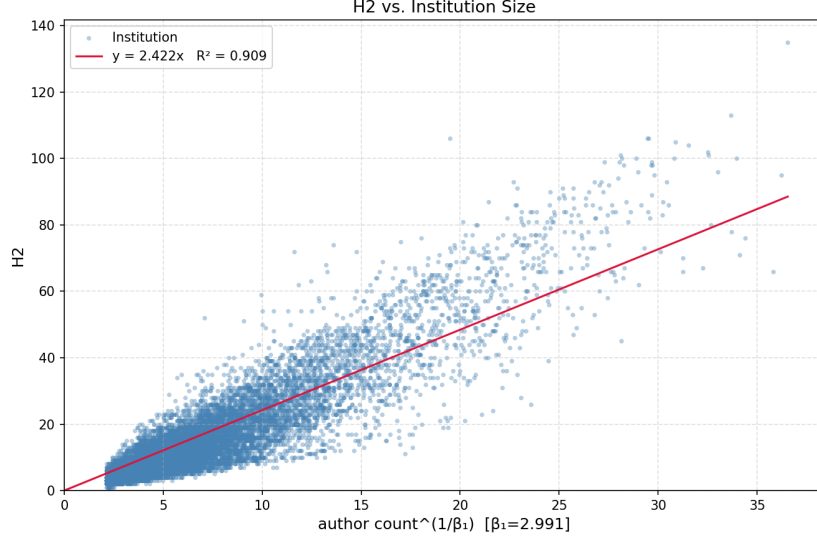
| Institution                        | Top 50 | Top 10 | Overall rank | Overall $h_2$ |
|------------------------------------|--------|--------|--------------|---------------|
| University of Oxford               | 19     | 5      | 6            | 105           |
| Harvard University                 | 18     | 12     | 1            | 135           |
| University of Michigan             | 18     | 7      | 22           | 96            |
| Stanford University                | 17     | 13     | 2            | 113           |
| University of Toronto              | 17     | 5      | 13           | 100           |
| University of California, Berkeley | 16     | 4      | 16           | 99            |
| University of Cambridge            | 16     | 3      | 5            | 106           |
| The University of Melbourne        | 16     | 1      | 31           | 92            |
| University College London          | 15     | 8      | 10           | 101           |

Oxford leads on top-50 breadth (19 of 26 fields), while Stanford leads on outright field leadership with 13 top-10 placements, narrowly ahead of Harvard’s 12. Five of the nine broadest institutions (Oxford, Toronto, Berkeley, Cambridge, and Melbourne) place fewer than half their top-50 fields in the top 10, meaning most of their breadth reflects consistent competence across many disciplines rather than dominance of any one. The University of Melbourne (rank 31,  $h_2 = 92$ ) is the only institution outside North America and the United Kingdom in this group. The contrast between breadth and depth is illustrated by comparing University College London ( $h_2 = 101$ , 15 top-50 fields) with Oxford ( $h_2 = 105$ , 19 top-50 fields): UCL leads 8 fields outright to Oxford’s 5, despite placing in fewer top-50 fields overall. At the other extreme, Imperial College London (rank 21,  $h_2 = 96$ ) achieves its overall position with zero top-10 field placements across 8 top-50 fields, through consistent mid-table performance rather than any single-field depth.

### 3.9 Institution efficiency

#### Model validation

Figure 1 plots institution-level  $h_2$  against the Egghe-predicted value  $S^{1/\beta_1}$  for all institutions with at least 10 OpenAlex authors. A no-intercept regression, appropriate because  $N = 0$  authors forces  $h_2 = 0$ , gives slope  $c = 2.422$  and uncentered  $R^2 = 0.909$ :  $S^{1/\beta_1}$  alone explains 91% of the variance in  $h_2$ , confirming that Egghe’s power-law functional form describes the data. The efficiency metric  $\varepsilon_2 = h_2/S^{1/\beta_1}$  captures each institution’s deviation from this overall trend.



**Fig. 1** Institution-level  $h_2$  versus  $S^{1/\beta_1}$  (Egghe’s predicted scaling,  $\beta_1 = 2.991$ ). No-intercept regression:  $h_2 = 2.422 S^{1/\beta_1}$ ,  $R^2 = 0.909$  (uncentered).

The cross-check on Egghe’s compounding structure fails sharply, however. Egghe’s derivation requires that the Lotka tail exponent of the  $h_1$  distribution equals the product  $\alpha_0\alpha_1$ ; these separate fits give  $\alpha_0 = 2.454$  and  $\alpha_1 = 2.994$ , whose product is  $7.35$ , against the directly-measured  $\beta_1 = 2.991$ : a discrepancy of  $146\%$ . The analogous prediction at the next level ( $\beta_2 = \beta_1\alpha_2$ ) is similarly violated:  $2.991 \times 1.987 = 5.944$  against the directly-measured  $\beta_2 = 2.847$  ( $109\%$  discrepancy).

The failure of the compounding formula does not, however, undermine the efficiency metric. The two claims play different roles. The compounding formula ( $\beta_1 = \alpha_0\alpha_1$ ) is a *theoretical prediction* about the relationship between separately-fitted exponents; it is tested here and found wanting. The power-law scaling  $h_2 \propto S^{1/\beta_1}$  is an *empirical finding* ( $R^2 = 0.909$ ) that does not depend on the compounding formula being correct.  $\beta_1$  is estimated directly from the distribution of  $h_1$  values across authors, not derived from  $\alpha_0\alpha_1$ , so the  $146\%$  discrepancy in the compounding formula has

no bearing on the exponent actually used in  $\varepsilon_2$ . This is checked against the [Scimago Institutions Rankings](#) later in this subsection.

### Efficiency rankings

The efficiency metric  $\varepsilon_2 = h_2/S^{1/\beta_1}$  addresses Mryglod et al.’s (2022) concern: because  $S^{1/\beta_1}$  accounts for 91% of the variance in  $h_2$ , dividing it out isolates the residual variation that is not explained by headcount alone.

Table 8 lists the top 10 institutions by  $\varepsilon_2$  among the [8,118](#) with at least 100 authors. The top of the ranking is dominated by small and highly specialized institutions. Focused research centers and specialized colleges recur throughout the top 20.

**Table 8** Top 10 institutions by efficiency  $\varepsilon_2 = h_2/S^{1/\beta_1}$  ( $\beta_1 = 2.991$ ) among those with at least 100 authors ( $n = 8,118$ ).

| Institution                                       | $\varepsilon_2$ | $h_2$ | Authors | $h_2$ rank |
|---------------------------------------------------|-----------------|-------|---------|------------|
| University of Minnesota Rochester                 | 7.35            | 52    | 348     | 503        |
| Kettering University                              | 6.19            | 72    | 1,539   | 150        |
| Académie Nationale de Médecine                    | 5.97            | 34    | 182     | 1,287      |
| Center for Astrophysics Harvard & Smithsonian     | 5.93            | 59    | 963     | 339        |
| Saint Camillus International Univ. of Health Sci. | 5.91            | 35    | 204     | 1,253      |
| Université Bourgogne Franche-Comté                | 5.63            | 51    | 730     | 525        |
| Stockholm Resilience Centre                       | 5.48            | 29    | 146     | 1,806      |
| University of Antwerp                             | 5.44            | 106   | 7,207   | 3          |
| Barcelona Institute of Science and Technology     | 5.44            | 74    | 2,462   | 125        |
| Rush University                                   | 5.40            | 54    | 980     | 451        |

The high-efficiency entries are a mix of specialized research institutes (Center for Astrophysics Harvard & Smithsonian, Stockholm Resilience Centre, Barcelona Institute of Science and Technology), a learned society (Académie Nationale de Médecine), medical and health sciences institutions (University of Minnesota Rochester, Saint Camillus), and a comprehensive research university (University of Antwerp). A systematic audit of disambiguation artifacts is beyond the scope of this paper, and the top of this table should be read with appropriate caution.

Table 9 restricts to the [629](#) institutions with at least 5,000 authors. The University of Antwerp leads outright ( $\varepsilon_2 = 5.44$ ,  $h_2 = 106$ ,  $h_2$  rank 3 among all  $\geq 100$ -author institutions), placing simultaneously in the top 5 on both raw  $h_2$  and efficiency within this subset. The Weizmann Institute of Science ( $\varepsilon_2 = 4.22$ ), Caltech ( $\varepsilon_2 = 4.10$ , efficiency rank 3), Princeton ( $\varepsilon_2 = 4.06$ ), and Erasmus University Rotterdam ( $\varepsilon_2 = 4.02$ ) follow, all achieving high efficiency relative to their size. Among the elite comprehensive research universities that anchor the raw  $h_2$  top 10, Caltech (efficiency rank 3 in this subset), the University of Geneva (rank 9), and Université Paris-Saclay (rank 69) are the most efficient, each achieving  $h_2 \geq 93$  with under 25,000 authors.

**Table 9** Top 10 institutions by efficiency  $\varepsilon_2 = h_2/S^{1/\beta_1}$  ( $\beta_1 = 2.991$ ) among those with at least 5,000 authors ( $n = 629$ ).

| Institution                                    | $\varepsilon_2$ | $h_2$ | Authors | $h_2$ rank |
|------------------------------------------------|-----------------|-------|---------|------------|
| University of Antwerp                          | 5.44            | 106   | 7,207   | 3          |
| Weizmann Institute of Science                  | 4.22            | 76    | 5,711   | 117        |
| California Institute of Technology             | 4.10            | 93    | 11,348  | 26         |
| Princeton University                           | 4.06            | 87    | 9,591   | 45         |
| Erasmus University Rotterdam                   | 4.02            | 81    | 7,943   | 79         |
| The University of Adelaide                     | 3.97            | 91    | 11,671  | 34         |
| University of Colorado Anschutz Medical Campus | 3.85            | 67    | 5,146   | 208        |
| Vrije Universiteit Amsterdam                   | 3.85            | 80    | 8,757   | 92         |
| University of Geneva                           | 3.83            | 80    | 8,847   | 91         |
| City University of Hong Kong                   | 3.81            | 70    | 6,062   | 167        |

### External validation against Scimago Institutions Rankings

Joining the Scopus-derived Scimago Institutions Rankings (SIR) 2026 Overall Rank export to Table 8 yields 3,571 matched institutions: 33.0% of SIR’s list and 44.0% of the paper’s efficiency table.

On the matched set, the Spearman rank correlation between raw  $h_2$  and SIR’s Global Rank is  $\rho = 0.798$  ( $p \approx 0$ ); between  $\varepsilon_2$  and SIR’s Global Rank it is  $\rho = 0.443$  ( $p \approx 10^{-191}$ ). Restricting the match to SIR’s “Universities” sector only (5,491 of SIR’s 10,827 institutions, excluding ministries, hospitals, and companies that SIR also ranks) gives essentially the same result ( $\rho = 0.802$  raw,  $\rho = 0.441$  efficiency,  $n = 3,515$ ), so the gap between the two correlations is not an artefact of comparing an academic  $h_2$  ranking against a mixed-sector SIR list.

Against SIR’s composite Global Rank, a source with no shared data infrastructure with OpenAlex, raw  $h_2$  tracks SIR’s Global Rank substantially more closely than  $\varepsilon_2$  does. I report both comparisons as description rather than as validation of the efficiency normalization; a plausible mechanistic reading of the SIR result is that SIR’s Global Rank is a composite that rewards institutional scale and visibility in ways that raw  $h_2$  partially captures and that dividing out size by construction removes.

### 3.10 $h_3$ : country level

Aggregating institutional  $h_2$  up to the country level via the same rank-based procedure yields  $h_3$ . Table 10 lists the ten highest- $h_3$  countries, and Figure 2 maps  $h_3$  for all 214 countries with at least one qualifying institution.

The United States leads by a wide margin, but Italy’s 44 on only 192 institutions and Japan’s 44 on 1,145 are a first hint that raw  $h_3$  rewards national research-system size as well as depth.

### 3.11 Country efficiency

Figure 3 repeats the size-scaling analysis at the country level, plotting  $h_3$  against  $R^{1/\beta_2}$  for each country. The no-intercept fit gives slope  $c = 3.534$  and uncentered  $R^2 = 0.900$ : institution count alone explains 90% of the variance in  $h_3$  across countries.

**Table 10** Top 10 countries by  $h_3$ .

| Country        | $h_3$ | Institutions |
|----------------|-------|--------------|
| United States  | 68    | 3,652        |
| China          | 53    | 1,071        |
| Germany        | 47    | 469          |
| United Kingdom | 46    | 472          |
| Italy          | 44    | 192          |
| Japan          | 44    | 1,145        |
| France         | 40    | 374          |
| Spain          | 38    | 140          |
| Australia      | 36    | 115          |
| South Korea    | 35    | 332          |

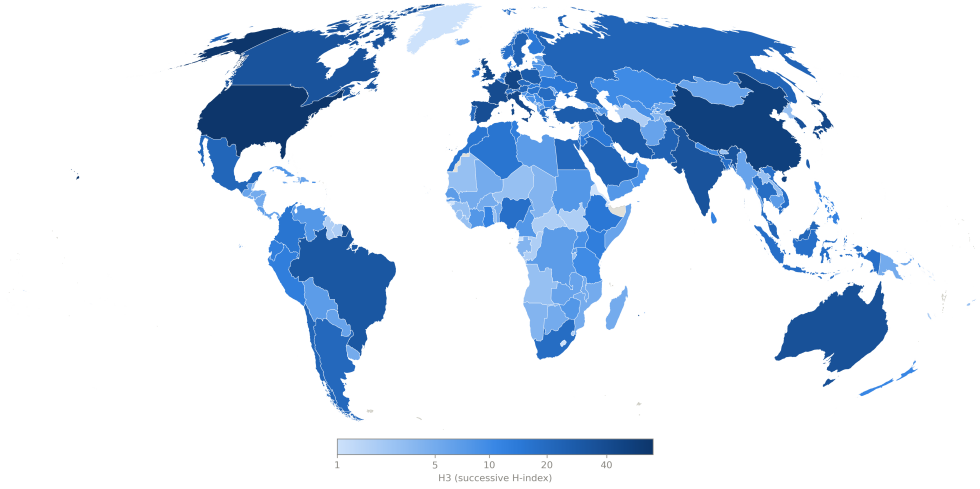
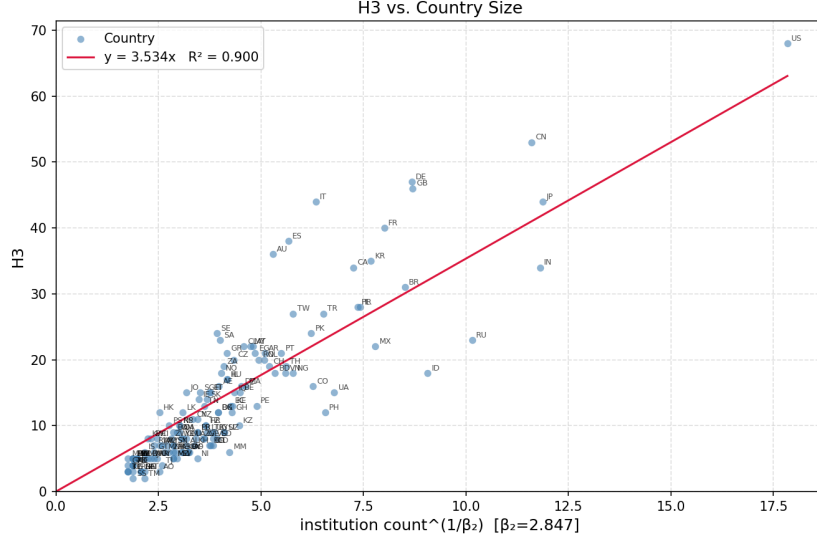
**Fig. 2**  $h_3$  by country. Countries with no qualifying institution are shown in grey.

Table 11 ranks the 151 countries with at least 5 qualifying institutions by efficiency  $\varepsilon_3 = h_3/R^{1/\beta_2}$ . The Italy–Japan contrast foreshadowed in Subsection 3.10 is quantified: Italy ( $\varepsilon_3 = 6.94$ , rank 1) achieves  $h_3 = 44$  from only 192 institutions, while Japan (efficiency rank 41,  $\varepsilon_3 = 3.71$ ) reaches  $h_3 = 44$  from 1,145 institutions, the same headline number but at  $6\times$  the institutional scale. Australia (rank 2,  $\varepsilon_3 = 6.80$ , 115 institutions) and Spain (rank 3,  $\varepsilon_3 = 6.70$ , 140 institutions) round out a top tier of countries with compact research systems that place well on this metric. Germany (rank 6), the United Kingdom (rank 7), and France (rank 8) are the only countries that place in the top 10 on both raw  $h_3$  and efficiency. Saudi Arabia is a notable outlier at 5th by efficiency ( $h_3=23$  on just 52 institutions,  $\varepsilon_3 = 5.74$ ) despite ranking only 21st by raw  $h_3$ ; as flagged in Subsection 2.8, this is plausibly an artifact of high-h-index researchers listing Saudi institutions as secondary affiliations in exchange for



**Fig. 3** Country-level  $h_3$  versus  $R^{1/\beta_2}$  (Egghe’s predicted scaling,  $\beta_2 = 2.847$ ). No-intercept regression:  $h_3 = 3.534 R^{1/\beta_2}$ ,  $R^2 = 0.900$  (uncentered).

funding (Bhattacharjee, 2011) rather than a genuine concentration of resident research talent.

The United States falls to efficiency rank 37 ( $\varepsilon_3 = 3.812$ ) despite its raw lead ( $h_3 = 68$ , rank 1). With 3,652 qualifying institutions, nearly 8× more than China and 19× more than Italy, even a large absolute  $h_3$  advantage is absorbed by the size denominator. China (rank 19,  $\varepsilon_3 = 4.571$ , 1,071 institutions) is more compact and appears above the US on this metric.

**Table 11** Top 10 countries by efficiency  
 $\varepsilon_3 = h_3/R^{1/\beta_2}$  ( $\beta_2 = 2.847$ ) among those with at least  
5 institutions ( $n = 150$ ).

| Country        | $\varepsilon_3$ | $h_3$ | Institutions | $h_3$ rank |
|----------------|-----------------|-------|--------------|------------|
| Italy          | 6.94            | 44    | 192          | 5          |
| Australia      | 6.80            | 36    | 115          | 9          |
| Spain          | 6.70            | 38    | 140          | 8          |
| Sweden         | 6.12            | 24    | 49           | 19         |
| Saudi Arabia   | 5.74            | 23    | 52           | 21         |
| Germany        | 5.42            | 47    | 469          | 3          |
| United Kingdom | 5.29            | 46    | 472          | 4          |
| Greece         | 5.04            | 21    | 58           | 28         |
| France         | 4.99            | 40    | 374          | 7          |
| Chile          | 4.81            | 22    | 76           | 23         |

### 3.12 Sensitivity to disambiguation error

Table 12 reports, for seven simulated population-wide error rates spanning  $p = 1\%$  to a deliberately implausible  $p = 80\%$  (Subsection 2.9), the Spearman rank correlation and top-set overlap between the perturbed and unperturbed rankings, averaged over 50 trials, together with how often the paper’s three headline field-leadership claims survive the perturbation intact. The cross-validation sample’s own raw outlier fraction ( $4/30 \approx 13\%$  across both waves, Subsection 3.1) is included as an extra row for reference, though it carries no special status in this analysis.

Institution- and country-level rank correlations decline gradually with  $p$ : Spearman  $\rho(h_2)$  falls from 0.9995 at  $p = 1\%$  to 0.9793 at  $p = 80\%$ ;  $\rho(h_3)$  from 0.9998 to 0.9946. This follows from how  $h_2$  and  $h_3$  are computed: each is an order statistic, so what matters for a given institution or country is whether a noisy author sits near its own threshold, not how many noisy authors exist elsewhere; and because the perturbation is applied simultaneously to all institutions, individual rank shifts are dampened at the aggregate level. Top-20 institution overlap follows a U-shaped pattern: 95% at  $p = 1\%$ , declining to a minimum of 44% near  $p = 20\%$ , then recovering to 83% at  $p = 80\%$ . This reflects a cancellation effect: at very high error rates, roughly equal numbers of inflation and deflation events accumulate within every institution’s author pool, partially restoring the pre-perturbation rank ordering among the top-ranked set.

The field-specific leadership claims differ sharply in sensitivity. UCAS’s Environmental Science lead survives all 50 trials at every error rate tested. Carnegie Mellon’s Computer Science lead is robust at  $p \leq 5\%$  (48/50 trials) but displaced at  $p = 10\%$  (6/50) and  $p = 13\%$  (1/50), recovering partially at  $p = 80\%$  (16/50) as the cancellation effect sets in. Wageningen’s Agricultural and Biological Sciences lead is the most sensitive: it holds 50/50 at  $p = 1\%$  but falls to 36/50 at  $p = 5\%$ , and to 0/50 at  $p = 10\%$  through  $p = 40\%$ , with partial recovery at  $p = 80\%$  (1/50). Wageningen’s greater sensitivity relative to CMU is consistent with the country-stratified error model: several of Wageningen’s main competitors in Agricultural and Biological Sciences are Chinese institutions, which receive  $3\times$  the base error rate and therefore a larger expected perturbation.

Institution and country rank correlations remain high across all tested rates ( $\rho(h_2) \geq 0.979$ ;  $\rho(h_3) \geq 0.994$ ), and the UCAS Environmental Science claim is similarly stable. The Wageningen Agricultural and Biological Sciences and Carnegie Mellon Computer Science field-leadership claims are sensitive above  $p \approx 5\%$  and should be read with correspondingly greater caution as single-institution results. The overall finding is that  $h_2$  and  $h_3$  rankings are structurally determined at population scale. Individual author errors are absorbed across many threshold comparisons, and at very high error rates the bidirectional perturbations increasingly cancel within institutions. This stability under the specific perturbation model tested cannot rule out concentrated or directional bias of the kind that would arise from systematic disambiguation errors targeting a single institution or country; that remains an open limitation.

**Table 12** Sensitivity of  $h_2$  and  $h_3$  rankings to simulated disambiguation error (50 trials per rate; bidirectional perturbation with country-stratified rates for CN/KR/IR/SA; see Subsection 2.9). Overlap is computed on the top 20 institutions (by  $h_2$ ) and top 10 countries (by  $h_3$ ). The last three columns give the fraction of trials in which the named institution keeps rank #1 in the stated field.

| $p$  | $\rho(h_2)$ | Top-20 | $\rho(h_3)$ | Top-10 | Wag. Ag&Bio #1 | UCAS EnvSci #1 | CMU CS #1 |
|------|-------------|--------|-------------|--------|----------------|----------------|-----------|
| 1%   | 0.9995      | 95%    | 0.9998      | 99%    | 50/50          | 50/50          | 50/50     |
| 5%   | 0.9977      | 78%    | 0.9992      | 99%    | 36/50          | 50/50          | 48/50     |
| 10%  | 0.9958      | 54%    | 0.9986      | 97%    | 0/50           | 50/50          | 6/50      |
| 13%* | 0.9946      | 47%    | 0.9983      | 96%    | 0/50           | 50/50          | 1/50      |
| 20%  | 0.9922      | 44%    | 0.9977      | 92%    | 0/50           | 50/50          | 0/50      |
| 40%  | 0.9865      | 56%    | 0.9965      | 90%    | 0/50           | 50/50          | 0/50      |
| 80%  | 0.9793      | 83%    | 0.9946      | 90%    | 1/50           | 50/50          | 16/50     |

\*cross-validation sample’s raw outlier fraction, included for reference only.

## 4 Conclusion

Schubert closed his 2007 note wondering whether  $h_2$  of institutes would prove more interesting than  $h_2$  of journals, absent the researcher–institute–country data needed to check. Nineteen years and 13.3 million OpenAlex authors later, the answer is yes, but not in the way a single top-10 table would suggest.

The overall  $h_2$  ranking is the least interesting object in this paper. Because Medicine alone accounts for 24% of the author pool, the unrestricted university-wide aggregate is, in substance, a ranking of medical school size. The field decomposition reveals a completely different pattern of field-level  $h_2$  rankings: Wageningen ranks first in Agricultural & Biological Sciences by field  $h_2$  while ranking 148th overall; the University of Chinese Academy of Sciences (UCAS) ranks first in Environmental Science, Materials Science, and Engineering by field  $h_2$  from 24th overall, though these leads hold only under the paper’s single-field assignment: UCAS leads Environmental Science at all multi-field thresholds tested, but CMU’s Computer Science lead falls to Stanford under the more aggressive 20% threshold (Subsection 3.4); Carnegie Mellon ranks first in Computer Science by field  $h_2$  from 159th under the single-field assignment; the University of Science and Technology of China ranks first in Energy from 66th; and Shenyang Pharmaceutical University ranks first in Pharmacology from 957th. Brazil offers perhaps the sharpest illustration of what the overall ranking hides: the Universidade de São Paulo and the Universidade Estadual de Campinas rank among the top six globally in Dentistry by field  $h_2$  despite neither country nor either institution appearing anywhere near the top of any general-purpose ranking. The field-level  $h_2$  is the unit that should be the default for comparison, not the aggregate.

On the methodological side, Egghe’s power-law functional form is empirically well-supported: author count alone explains 91% of the variance in  $h_2$ , and institution count alone explains 90% of the variance in  $h_3$ . However, the theoretical compounding prediction  $\alpha_0\alpha_1 \approx \beta_1$  fails by 146%. The model is empirically useful; the underlying theory needs a different reading.

Against Scimago Institutions Rankings 2026 (SCImago Research Group, 2026), a Scopus-derived ranking independent of OpenAlex, raw  $h_2$  correlates at  $\rho = 0.798$  against SIR’s Global Rank versus  $\rho = 0.443$  for  $\varepsilon_2$ , on a smaller, name-matched set of

3,571 institutions (Subsections 2.12 and 3.9). Against SIR’s composite Global Rank, raw  $h_2$  tracks the external ranking substantially more closely than  $\varepsilon_2$  does, plausibly because SIR’s composite is itself influenced by scale-correlated factors that dividing out size removes; this is reported as a description of the metrics’ behaviour, not as validation of the efficiency normalisation’s superiority.

Several limitations constrain interpretation. Author disambiguation is the most consequential open problem: the four outlier cases identified in cross-validation (notably Lei Jiang,  $h_1^{\text{OpenAlex}} = 215$  against Web of Science = 115) confirm that merger errors cluster in non-Anglophone and common-name strata, exactly the authors whose inflated  $h_1$  values would propagate most damagingly into institutional and country-level rankings. That cross-validation sample was deliberately adversarial, however, and cannot by itself say how much of the full 13.3-million-author population is affected; a Monte Carlo sensitivity analysis run at population-wide error rates from 1% to a deliberately implausible 80% (Subsections 2.9, 3.12) finds institution- and country-level rank correlations declining gradually but remaining high across all tested rates ( $\rho(h_2) \geq 0.979$ ;  $\rho(h_3) \geq 0.994$ ); the UCAS Environmental Science claim is stable throughout, while the Wageningen Agricultural and Biological Sciences and Carnegie Mellon Computer Science field-leadership claims are sensitive above  $p \approx 5\%$  and should be read with correspondingly greater caution. Arts and Humanities are systematically under-covered in all major citation databases: both threshold authors sampled at Oxford returned no value in Web of Science, Scopus, or Google Scholar, and h-index comparisons in book-based fields should be read with correspondingly less confidence. Field-specific rankings are also sensitive to how field membership is defined: a multi-field assignment (authors counted in every field comprising  $\geq 33\%$  of their output) yields highly stable overall rankings ( $\rho = 0.960$ ); UCAS retains its Environmental Science lead at all multi-field thresholds tested, while Carnegie Mellon’s Computer Science lead falls to Stanford at the more aggressive 20% threshold (Subsection 3.4); these findings confirm that the field-leadership claims should be read as reflecting depth of specialisation rather than unconditional supremacy. OpenAlex’s inconsistent retraction flagging leaves a small, unquantified fraction of  $h_1$  values potentially inflated. Saudi Arabia’s efficiency rank (5th,  $\varepsilon_3 = 5.74$ ) is plausibly an artifact of affiliation-listing for funding rather than resident research presence (Bhattacharjee, 2011), but Saudi Arabia is simply the case where this kind of distortion is most visible, not necessarily the only country affected. Finally,  $h_2$  is not comparable across fields of different size: Decision Sciences (max  $h_2 = 20$ ) cannot be placed on the same scale as Medicine (max  $h_2 = 118$ ), and the results should be read within fields only.

The most impactful near-term improvement to this framework would require a data source that does not yet exist: departmental affiliation in OpenAlex. If OpenAlex were to record the department within an institution alongside the institution itself, the hierarchy would extend naturally to researcher  $\rightarrow$  department  $\rightarrow$  institution  $\rightarrow$  country, with successive indices  $h_1$  through  $h_4$ . Departments are field-specialized by construction, so the step of inferring each author’s modal field from topic weights may become unnecessary at the departmental level. I flag this as the single most valuable addition OpenAlex could make for scientometric work of this kind.

## Statements and Declarations

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